

# YEAR 9

## End of Year Assessment

There are 65 marks on this paper.

You have 60 minutes to complete the paper.

Write in black ink.

You can use a calculator and ruler.

A COPY OF THE PERIODIC TABLE AND PHYSICS EQUATIONS CAN BE FOUND  
AT THE END OF THIS QUESTION PAPER.

|               |                                           |
|---------------|-------------------------------------------|
| Forename      | Tejas                                     |
| Surname       | Dilip                                     |
| Class/Teacher | <del>Mrs Chinn / Akshindate / Tiger</del> |

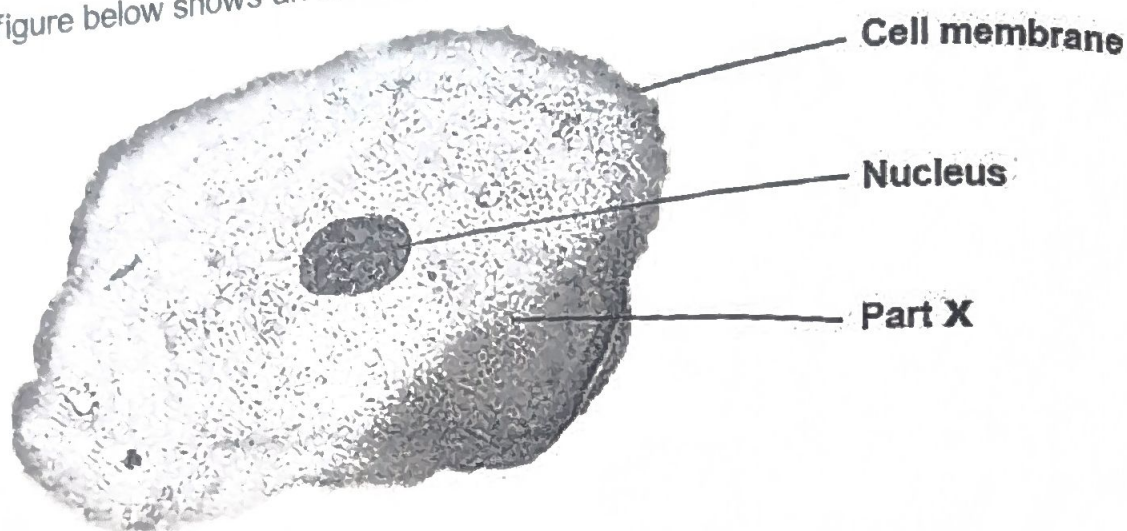
9GB/G2

# BIOLOGY (6)

Q1.

This question is about cells.

(a) The figure below shows an animal cell seen through a microscope.



What is part X in the figure above?

Tick (✓) **one** box.

Cytoplasm

☒

Mitochondrion

☐

Vacuole

☐

(b) Name **one** structure found in the nucleus.

DNA

(c) Name **one** cell part that is found in a plant cell but is **not** found in an animal cell.

~~vacuole~~ chloroplast

- (d) Root hair cells are found in plants.

Which substance do root hair cells absorb from the soil?

Tick (✓) **one** box.

|         |                                     |
|---------|-------------------------------------|
| Glucose | <input type="checkbox"/>            |
| Protein | <input type="checkbox"/>            |
| Water   | <input checked="" type="checkbox"/> |

(1)

- (e) A sperm cell is a specialised cell.

What is the function of a sperm cell?

To find and enter the egg cell  
so fertilisation can happen

(1)

- (f) Draw **one** line from each specialised cell to the type of organism where the specialised cell is found.

| Specialised cell | Type of organism |
|------------------|------------------|
| Nerve cell       | Animals          |
|                  | Bacteria         |
| Xylem cell       | Fungi            |
|                  | Plants           |

(2)

(g) The cell membrane controls which substances move into and out of cells.  
Which **two** processes move substances into and out of cells?

Tick (✓) **two** boxes.

Active transport

☒

Diffusion

☒

Mitosis

☐

Mutation

☐

Respiration

☒

(h) The table below shows substances that are exchanged between the blood and muscle cells.

Complete the table below to show the direction of movement of each substance.

Tick (✓) **one** box in each row.

| Substance      | Direction of movement of substance  |                                     |                                                           |
|----------------|-------------------------------------|-------------------------------------|-----------------------------------------------------------|
|                | ONLY from blood to muscle cells     | ONLY from muscle cells to blood     | From blood to muscle cells AND from muscle cells to blood |
| Carbon dioxide |                                     | <input checked="" type="checkbox"/> |                                                           |
| Oxygen         | <input checked="" type="checkbox"/> |                                     | <input checked="" type="checkbox"/>                       |
| Water          |                                     |                                     | <input checked="" type="checkbox"/>                       |

(3)  
(Total 12 marks)



**Q2.**

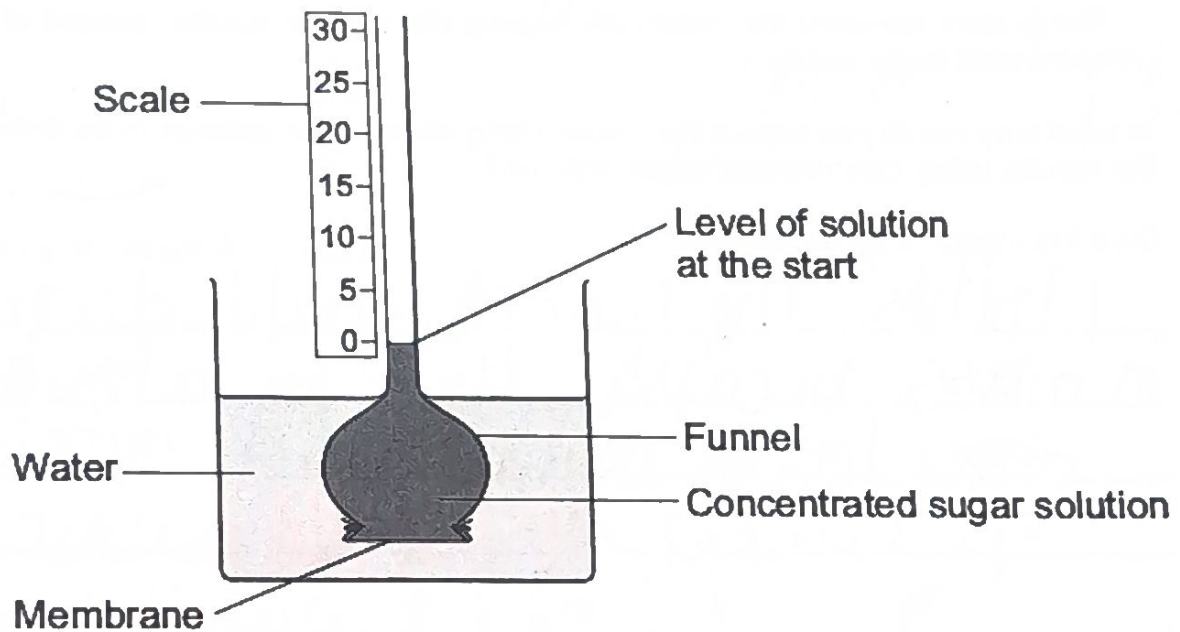
Some substances move through membranes.

A student set up an investigation.

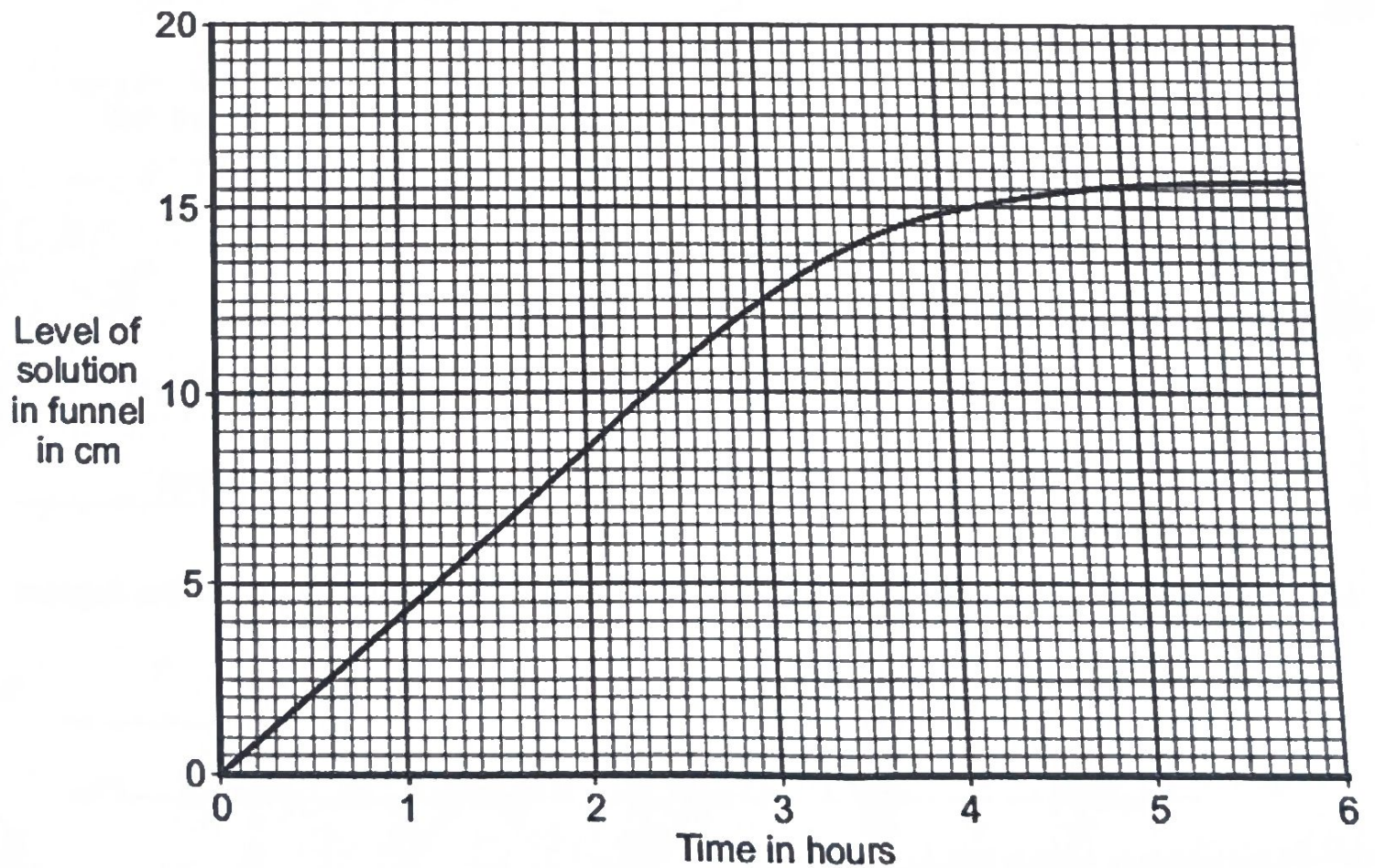
The student:

- tied a thin membrane across the end of a funnel
- put concentrated sugar solution in the funnel
- put the funnel in a beaker of water
- measured the level of the solution in the funnel every 30 minutes.

The diagram shows the apparatus.



The graph shows the results.





- (a) After 3 hours, the level of the solution in the funnel is different from the level at the start.

Explain why, as fully as you can.

The water moved into the funnel because of osmosis. The funnel had a concentrated sugar solution and a lower water concentration and the funnel also had a thin membrane causing the water from the beaker to move from high to low concentration.

- (b) The student repeated the investigation using dilute sugar solution instead of concentrated sugar solution.

In what way would you expect the results using dilute sugar solution to be different from the results using concentrated sugar solution?

Give the reason for your answer.

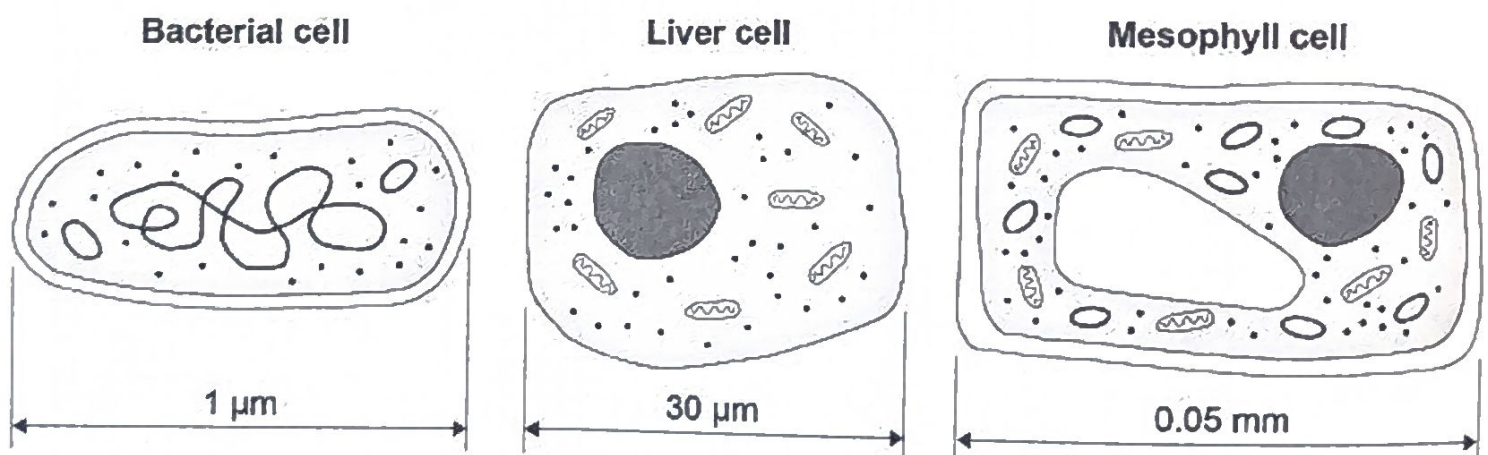
I think the funnel would decrease in mass because there is a higher ~~the~~ lower concentration outside the funnel.

It would increase slower not decrease

(Total 5 marks)

Q3.

The diagram below shows three types of cell.



- (a) Give **two** similarities between the prokaryotic cell and the eukaryotic cells in the diagram above.

1 Ribosomes

2 ~~Cell membrane~~ Cell membrane

- (b) Give **three** differences between the prokaryotic cell and the eukaryotic cells in the diagram above.

- 1 Loose DNA Bacterial cell has loose DNA
- 2 Bacterial cells significantly smaller
- 3 Membrane bound nucleus for prokaryotic which!

(3)

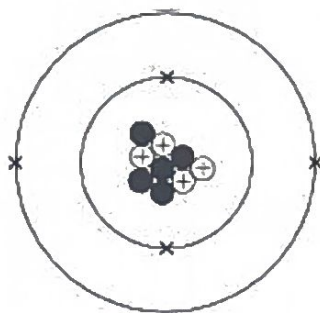
## CHEMISTRY

### Q1.

All substances are made from atoms.

- (a) **Figure 1** represents a beryllium atom.

**Figure 1**



What is the number of protons and the number of neutrons in the beryllium atom?

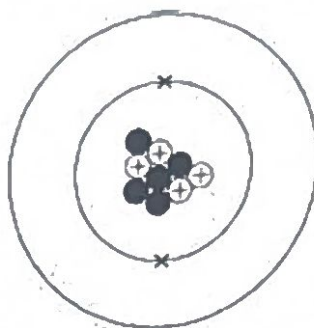
Number of protons 4

Number of neutrons 5

(2)

(b) Figure 2 represents a beryllium ion.

Figure 2



What is the relative charge on a beryllium ion?

Tick (✓) **one** box.

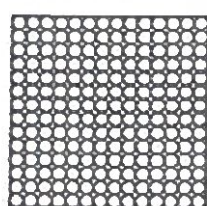
0 ☐

+1 ☒

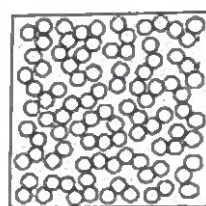
+2 ☐

Figure 3 shows the arrangement of atoms in the three states of matter.

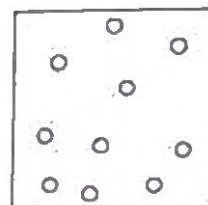
Figure 3



A



B



C

(c) What state of matter is represented by state C in Figure 3?

Tick (✓) **one** box.

Gas

☒

Liquid

☐

Solid

☐



- (d) What is the name of the process when state **B** changes into state **A**?

Use Figure 3.

Tick (✓) **one** box.

Condensing

☐

Freezing

☒

Melting

☐

(1)

- (e) How can state **B** be changed into state **C**?

Use Figure 3.

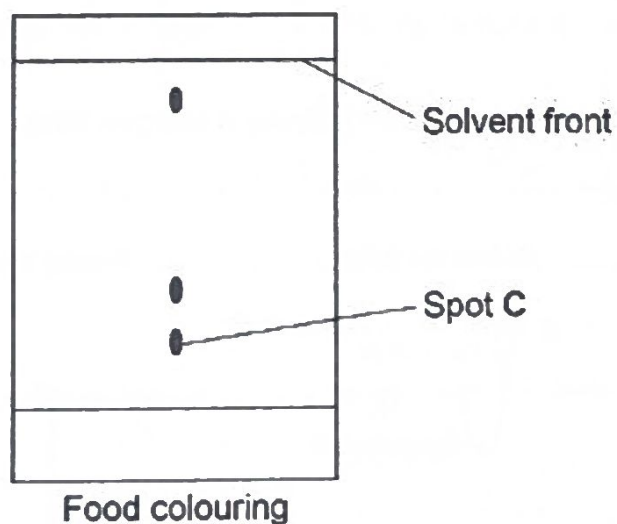
Evaporation

(1)

(Total 6 marks)

## Q2.

The diagram shows a chromatogram for a food colouring.



- (a) How does the chromatogram show that the food colouring is a mixture?

It shows three different spots and they are all different colours which means it is a mixture

(1)

- (b) A student makes measurements for spot C.

The table shows the results.

|                           | Distance in mm |
|---------------------------|----------------|
| Distance moved by spot C  | 7              |
| Distance moved by solvent | 39             |

Calculate the  $R_f$  value for spot C.

Give your answer to 2 significant figures.

Use the results in the table.

$$\frac{7}{39}$$

*distance moved by spot C / d of solvent*

$$R_f \text{ value} = 5.6$$

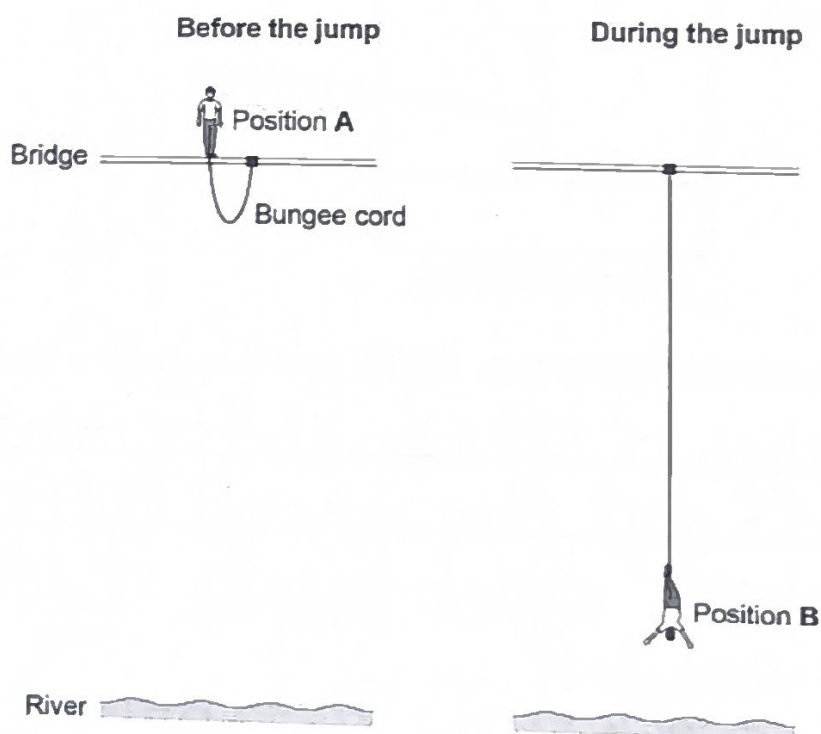
(3)

## PHYSICS

Q1.

The figure below shows a student before and during a bungee jump.

The diagram is not to scale.



- (a) In position **B**, the student is moving towards the river and the bungee cord is stretching. How do the energy stores in position **B** compare with the energy stores in position **A**?

Tick (✓) **one** box in **each** row.

| Energy store                                 | Less than at A | The same as at A | More than at A |
|----------------------------------------------|----------------|------------------|----------------|
| The student's gravitational potential energy | ✓              |                  |                |
| The student's kinetic energy                 |                |                  | ✓              |
| The bungee cord's elastic potential energy   |                |                  | ✓              |

(3)

- (b) The bungee cord behaves like a spring with a spring constant of 78.4 N/m.

At one point in the bungee jump, the extension of the bungee cord is 25 m.

Calculate the elastic potential energy stored by the bungee cord.

Use the equation:

$$\text{elastic potential energy} = 0.5 \times \text{spring constant} \times \text{extension}^2$$

$$0.5 \times 78.4 \times 25^2$$

$$\text{Elastic potential energy} = 24500 \text{ J}$$

(2)



The table below shows information about different bungee cords.

| Bungee cord | Spring constant in N/m | Maximum extension before snapping in metres |
|-------------|------------------------|---------------------------------------------|
| A           | 78.4                   | 36                                          |
| B           | 82.0                   | 24                                          |
| C           | 84.5                   | 12                                          |

- (c) Bungee cord **C** will have a smaller extension than **A** or **B** for any bungee jumper.

Give the reason why.

It has the smallest max extension before snapping

(1)

- (d) Which bungee cord would be safest to use for a person with a large weight?

Give a reason for your answer.

Bungee cord C

Reason Because It has the least max extension it will slow the person down before its too low as the spring could get damaged

(2)

(Total 8 marks)

Q2.



The image shows a battery-powered drone.

- (a) Complete the sentences.

Choose the answers from the box.

|                         |                   |
|-------------------------|-------------------|
| chemical                | elastic potential |
| gravitational potential | kinetic nuclear   |

As the drone accelerates upwards its gravitational potential energy increases and its kinetic energy increases. The chemical energy store of the battery decreases.

(3)

- (b) In the USA, drones are not allowed to be flown too high above the ground.

Suggest **one** possible risk of flying a drone too high above the ground.

~~It could injure birds~~  
 It could run out of battery  
 and could fall onto humans ~~or~~ animals injuring them

- (c) Write down the equation that links energy transferred, power and time.

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

- (d) The drone can fly for 25 minutes before the battery needs recharging. The power output of the battery is 65.0 W. Calculate the maximum energy stored by the battery.

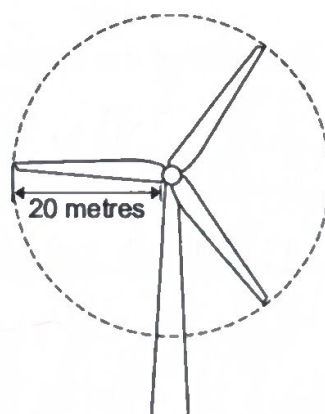
$$65 = \frac{E}{25} \Rightarrow E = 65 \times 25$$

$$\text{Maximum energy} = 1625 \text{ joules}$$

(3)  
(Total 8 marks)

### Q3.

The diagram shows a wind turbine.



- (a) The blades of the turbine are 20 metres long. On average, 15 000 kg of air, moving at a speed of 12 m/s, hit the blades every second.

Calculate the kinetic energy of the air hitting the blades every second.

Show clearly how you work out your answer.

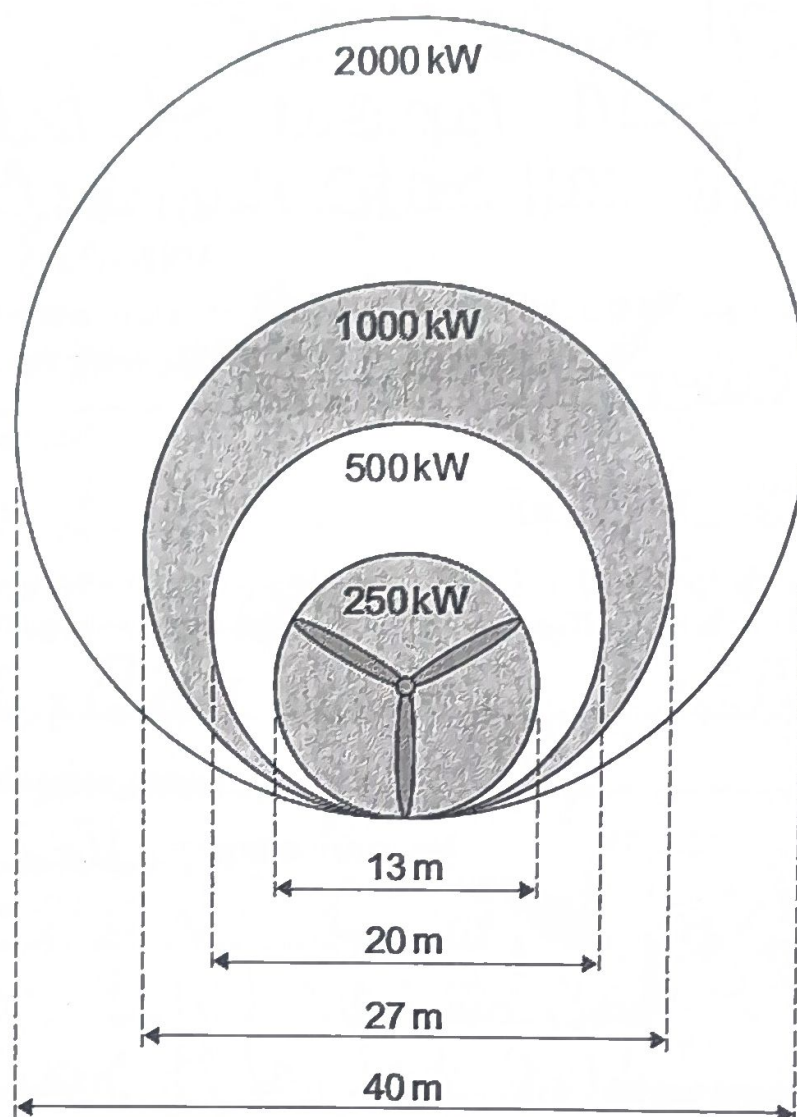
$$Ke = \frac{1}{2} \times 15000 \times (12)^2$$

$$\text{Kinetic energy} = 1080000 \text{ J}$$

(2)



- (b) Part of the kinetic energy of the wind is transformed into electrical energy. The diagram shows that, for the same wind speed, the power output of a turbine, in kilowatts, depends on the length of the turbine blades.



Give a reason why doubling the diameter of the blades more than doubles the power output of a turbine.

Because if we had an 20m turbine it would release 500kw  
 20 doubled is 40 and 40 releases 2000kw so it releases more than double

(Total 3 marks)



# chromatography

- (c) Plan a chromatography experiment to investigate the colours in an ink.

Firstly get a piece of paper, your ink a pencil a splint and a paperclip and a beaker filled with water. Not fully filled around a quarter then draw a line with pencil near the bottom of the paper, place a ~~drop~~ a drop of ink on the pencil line and hook the piece of paper onto the splint using a paperclip make sure the water is under the paper line. Wait until the ink stops moving up the page then calculate the  $R_f$  value by doing the  $\frac{\text{distance moved by spot}}{\text{distance of solvent}}$

(6)

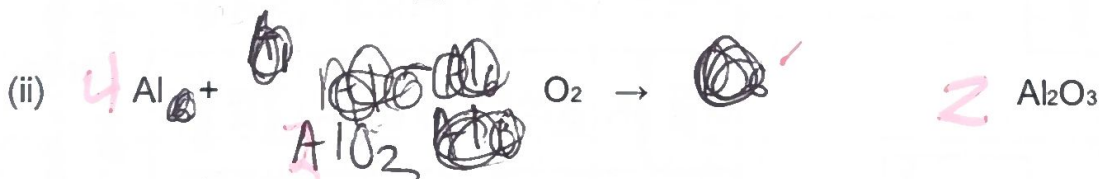
(Total 10 marks)

### Q3.

- (a) Balance these chemical equations.



(1)



(1)

- (b) Briefly explain why an unbalanced chemical equation cannot fully describe a reaction.

As if it is not balanced the chemical reaction will not work properly / not be accurate

(2)

(Total 4 marks)

**END OF QUESTIONS**

[illegible]

\* The Lanthanides (atomic numbers 58 – 71) and the Actinides (atomic numbers 90 – 103) have been omitted. Relative atomic masses for Cu and Cl have not been rounded to the nearest whole number.



| Equation number | Word equation                                                                                       | Symbol equation           |
|-----------------|-----------------------------------------------------------------------------------------------------|---------------------------|
| 1               | weight = mass $\times$ gravitational field strength ( $g$ )                                         | $W = m g$                 |
| 2               | work done = force $\times$ distance (along the line of action of the force)                         | $W = F s$                 |
| 3               | force applied to a spring = spring constant $\times$ extension                                      | $F = k e$                 |
| 4               | moment of a force = force $\times$ distance (normal to direction of force)                          | $M = F d$                 |
| 5               | pressure = $\frac{\text{force normal to a surface}}{\text{area of that surface}}$                   | $p = \frac{F}{A}$         |
| 6               | distance travelled = speed $\times$ time                                                            | $s = v t$                 |
| 7               | acceleration = $\frac{\text{change in velocity}}{\text{time taken}}$                                | $a = \frac{\Delta v}{t}$  |
| 8               | resultant force = mass $\times$ acceleration                                                        | $F = m a$                 |
| 9 HT            | momentum = mass $\times$ velocity                                                                   | $p = m v$                 |
| 10              | kinetic energy = $0.5 \times \text{mass} \times (\text{speed})^2$                                   | $E_k = \frac{1}{2} m v^2$ |
| 11              | gravitational potential energy = mass $\times$ gravitational field strength ( $g$ ) $\times$ height | $E_p = m g h$             |
| 12              | power = $\frac{\text{energy transferred}}{\text{time}}$                                             | $P = \frac{E}{t}$         |
| 13              | power = $\frac{\text{work done}}{\text{time}}$                                                      | $P = \frac{W}{t}$         |
| 14              | efficiency = $\frac{\text{useful output energy transfer}}{\text{total input energy transfer}}$      |                           |



| Equation number | Word equation                                                              | Symbol equation      |
|-----------------|----------------------------------------------------------------------------|----------------------|
| 15              | efficiency = $\frac{\text{useful power output}}{\text{total power input}}$ |                      |
| 16              | wave speed = frequency $\times$ wavelength                                 | $v = f \lambda$      |
| 17              | charge flow = current $\times$ time                                        | $Q = I t$            |
| 18              | potential difference = current $\times$ resistance                         | $V = I R$            |
| 19              | power = potential difference $\times$ current                              | $P = V I$            |
| 20              | power = (current) <sup>2</sup> $\times$ resistance                         | $P = I^2 R$          |
| 21              | energy transferred = power $\times$ time                                   | $E = P t$            |
| 22              | energy transferred = charge flow $\times$ potential difference             | $E = Q V$            |
| 23              | density = $\frac{\text{mass}}{\text{volume}}$                              | $\rho = \frac{m}{V}$ |

Students should be able to select and apply the following equations from the Physics equation sheet.

Equations required for Higher Tier papers only are indicated by HT in the left hand column.

| Equation number | Word equation                                                                                                              | Symbol equation                   |
|-----------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1 HT            | pressure due to a column of liquid = height of column $\times$ density of liquid $\times$ gravitational field strength (g) | $p = h \rho g$                    |
| 2               | (final velocity) <sup>2</sup> – (initial velocity) <sup>2</sup> = 2 $\times$ acceleration $\times$ distance                | $v^2 - u^2 = 2 a s$               |
| 3 HT            | force = $\frac{\text{change in momentum}}{\text{time taken}}$                                                              | $F = \frac{m \Delta v}{\Delta t}$ |
| 4               | elastic potential energy = 0.5 $\times$ spring constant $\times$ (extension) <sup>2</sup>                                  | $E_e = \frac{1}{2} k e^2$         |
| 5               | change in thermal energy = mass $\times$ specific heat capacity $\times$ temperature change                                | $\Delta E = m c \Delta \theta$    |
| 6               | period = $\frac{1}{\text{frequency}}$                                                                                      |                                   |
| 7               | magnification = $\frac{\text{image height}}{\text{object height}}$                                                         |                                   |